ECOTITANS

FOOD WASTE REDUCTION IN HOSTEL

Project by-

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PROBLEM STATEMENT-

In most hostels, food is prepared based on rough estimates rather than actual demand. This leads to:

- **Excess food preparation**
- **High food wastage**
- **Inefficient use of resources**
- **Increased operational costs**
- **Environmental impact due to wasted food**

Therefore, there is a need for a digital system that helps predict meal attendance and optimize food preparation.

OBJECTIVES:-

- To predict the number of students attending meals.
- To help kitchen staff prepare food according to demand.
- To track leftover food and analyze waste patterns.
- To encourage responsible food consumption among students.

PROPOSED SOLUTION:-

A smart food demand prediction system that includes a mobile application or website where students can confirm their meal attendance in advance.

The system will collect attendance data and use it along with other factors such as day of the week, holidays, and historical patterns to predict the expected number of students for each meal.

The predicted attendance will be displayed on a dashboard for the kitchen staff, helping them prepare the appropriate amount of food.

SYSTEM FEATURES:-

1. MEAL ATTENDANCE PREDICTION

Students can mark their availability for upcoming meals (breakfast, lunch, dinner) through the app or website

Features:-

- **Simple yes/no confirmation**
- **Ability to mark attendance for multiple days**
- **Easy interface for students**

IMPACTS:

- Helps estimate the number of students attending meals
- Reduces uncertainty for kitchen staff

2. AI-BASED DEMAND PREDICTION

The system will analyze:

- Historical attendance data
- Day of the week
- Holidays
- Exam periods
- Special events

Using this data, the system will generate a prediction of the expected number of students for each meal.

3. KITCHEN DASHBOARD

A dedicated dashboard will be available for mess staff.
The dashboard will show:-

- **Predicted student count**
- **Confirmed meal attendance**
- **Historical data**
- **Alerts if expected attendance changes significantly**

IMPACTS:

- Better planning for food preparation
- Efficient resource management

4. FOOD WASTE TRACKING

After each meal, the mess staff will enter the amount of leftover food.
The system will store:

- Type of meal
- Quantity of leftover food
- Date and time

IMPACTS:

- Helps identify patterns in food wastage
- Enables improvement in future predictions

5. ANALYTICS AND REPORTS

The system will generate reports such as:

- Weekly food consumption reports
- Monthly waste reports
- Attendance trends
- Waste reduction progress

These reports will help hostel management evaluate system effectiveness.

6. NOTIFICATIONS AND REMINDERS

- Students will receive notifications reminding them to confirm their meals.
- Examples:
  Reminder to confirm tomorrow's meals
  Notification if confirmation deadline is approaching

IMPACTS:

- Improves participation
- Makes predictions more accurate

7. SYSTEM ARCHITECTURE

Users of the System

1. Students

- Confirm meal attendance
- Receive reminders

2. Mess Staff

- View expected number of students
- Record leftover food

3. Admin/Hostel Management

- View reports and analytics
- Monitor waste reduction

TECHNOLOGY STACK UTILIZED-

FRONTEND DEVELOPMENT:
Implemented using HTML, CSS, and JavaScript to design an interactive and user-friendly interface.

BACKEND DEVELOPMENT:
Developed using Java or Python to handle application logic, data processing, and system operations.

DATABASE MANAGEMENT:
MySQL or Firebase is used for efficient storage and management of attendance records and food-related data.

ARTIFICIAL INTELLIGENCE / MACHINE LEARNING:
Basic machine learning algorithms are incorporated to analyze patterns and predict food demand.

PLATFORM:
Designed as a Web Application or Mobile Application for easy accessibility and scalability.

DEVELOPMENT TOOLS:
Tools such as Android Studio and Visual Studio Code (VS Code) are used for coding, testing, and deployment.

Conclusion -

The proposed Smart Food Demand Prediction and Management System aims to address the issue of food wastage in hostels through a combination of student participation, data analysis, and digital tools. By predicting meal demand and monitoring food waste, the system can help create a more efficient and sustainable hostel mess environment